PROJECT TECHNICAL DOCUMENTATION

BioSpecInfo

Web platform for computational chemistry, physics and mathematics

Technical Dossier

DOCUMENT CONTROL

Identifier	`BSI-DOC-00-EN`
Title	Technical Dossier
Software version	**bsi-v250**
Revision (commit)	`2c08fbc`
Date of issue	2026-11-12
Author	Samuele Pio Provenzano
Language	English
Status	Issued with version bsi-v250

Document generated from the versioned source in the project repository. The version and revision shown here and in the footer of every page identify exactly the state of the software described.

Contents

1. Executive summary

2. Product factsheet

3. Verified capabilities (current code state)

4. Technology stack

5. Author's declaration (self-attestation)

6. Attached documentation

BioSpecInfo — Technical Dossier

Open-access web platform for computational chemistry and the interactive consultation of chemical data on any device.

Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Profile	Cheminformatics Engineer / Scientific Software Developer
Software type	Progressive Web App (PWA), client-side, local-first, offline-first
Live demo	https://samupropio1-ship-it.github.io/BioSpecInfo-v11/
Repository	github.com/samupropio1-ship-it/BioSpecInfo-v11

> This dossier summarises the application and is the cover of the full technical documentation (SAD, Validation & Benchmark, Security & Compliance, Licensing) attached below.

1. Executive summary

BioSpecInfo is an **interactive cheminformatics platform** that runs all of its scientific logic **in the client browser**, with no server: molecular analysis, spectral prediction, 2D/3D modelling and educational tools — all **free, installable as an app and working offline**.

Its distinctive value is the **rare combination of advanced chemistry and modern software engineering** (WebAssembly, WebGL, PWA), with spectral predictors **quantitatively validated** against the official SDBS and NIST databases.

2. Product factsheet

Value proposition — A **≈4.5 MB** web platform (main file) that runs molecular analysis, spectral prediction and 3D visualisation at **zero server cost** and **100% offline-ready**.

Technical architecture	Content & capabilities
Client-side / Local-First (no backend)	**RDKit** cheminformatics engine (WebAssembly)
Single-file HTML5 / ES6, installable as **PWA**	**Spectra Centre**: IR, $^{1}H/^{13}C$-NMR, MS + isotope pattern, UV-Vis, Raman
2D (SVG/Canvas) and **3D** (three.js, 3Dmol.js) rendering	**Organic Chemistry**: 25 interactive modules
Service Worker offline-first, versioned	**Retrosynthesis**: 46 strategies + animated 2D/3D mechanisms

Technical architecture	Content & capabilities
Zero copyleft dependencies (MIT/BSD-3 only)	**3D Drug Atlas**: 63 diseases (23 tumours), 3D organs, PubMed
Local storage (localStorage, SQLite via sql.js)	3D astrochemistry, Quantum Lab, Academy (SM-2 quizzes)

Quantitative validation (thesis benchmark)

- **^{1}H-NMR** vs SDBS → **MAE 0.31 ppm** (89% of signals within ±0.5 ppm)
- **IR (C=O band)** vs NIST WebBook → **MAE 11 cm^{-1}** (90% within ±30 cm^{-1})
- Interactive performance **30–60 fps** from iPhone/iPad to Raspberry Pi 4

---

3. Verified capabilities (current code state)

Spectra Centre (molecular editor & analysis) — draw from scratch (full periodic table, templates, drag-to-bond) or load via SMILES/library →

- All chemical notations: formula, canonical SMILES, CXSMILES, SMARTS, InChI,

InChIKey, mol block, common/IUPAC name, CAS (one-click copy).

- Predicted spectra with bibliographic sources; **isotope pattern** (IUPAC

abundances) and MS **adduct masses**.

- Functional groups and properties (H donors/acceptors, aromatic, rotatable,

stereocentres) **highlightable on the 2D and 3D structure**.

- Drug-likeness: **Lipinski, Veber, Egan, Ghose, Muegge**, ESOL solubility,

QED, **Brenk/PAINS structural alerts**, physicochemical radar.

- **Tanimoto** similarity (Morgan ECFP4 fingerprint), elemental analysis, SMARTS

substructure search, SVG/PNG/MOL export.

Synthesis & Retrosynthesis — 46 strategies with an **animated mechanism player** in **2D and 3D**: *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

Organic Chemistry — 25 modules (fundamentals → heterocycles, pericyclic & Woodward–Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups).

Data Science — 3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), ML labs, SQL playground.

Other — 3D astrochemistry, Quantum Lab, Academy with spaced repetition (SM-2), interactive simulations, databases of elements/drugs/biomolecules.

> Overall metrics declared in the thesis (historical data assets): 324 organic reactions in vector SVG graphics, 26 biosynthetic pathways, 140+ drugs, 80+ biomolecules, 118 elements, 10 quantum simulators.

---

4. Technology stack

`HTML5` · `ES6/ESNext` · `WebAssembly` (RDKit MinimalLib, sql.js/SQLite) · `WebGL` (three.js, 3Dmol.js) · `SVG` / `Canvas 2D` · `Service Worker` / `PWA` · public API integration (PubChem PUG-REST, NIST, SDBS, MassBank, nmrshiftdb2, NASA/ESA).

---

5. Author's declaration (self-attestation)

The undersigned **Samuele Pio Provenzano** declares that:

1. The software **BioSpecInfo** is the author's **original work**, except for the

third-party open-source libraries (RDKit, 3Dmol.js, three.js, SmilesDrawer, sql.js) used under their respective permissive licenses and documented in the *Open-Source Licensing & IP Matrix*.

1. The reported validation data (MAE 0.31 ppm for ^{1}H-NMR vs SDBS; MAE 11 cm^{-1} for

the IR C=O band vs NIST) were measured by the author as part of the thesis work; the predictors are **heuristic group-additivity** models, suitable for educational use and qualitative functional-group identification, with the limitations stated in the *Validation Report*.

1. The architecture is **local-first**: no user personal data is transmitted to

author-controlled servers (alignment with GDPR principles).

> This is an **author's declaration**, not a certification issued by a third party.

---

6. Attached documentation

1. **Software Architecture Document (SAD)** — architecture, components, technologies.
2. **Verification & Validation & Benchmark Report** — test method, benchmarks, limitations.
3. **Security, Privacy & Compliance** — local-first, GDPR, OWASP, GAMP 5 (alignment).
4. **Open-Source Licensing & IP Matrix** — dependencies, licenses, intellectual property.

Software license: proprietary — "All rights reserved" (see `LICENSE`).